

Wave 5.1 Harmonic-Prior Residual Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

The first real wave 5.1 harmonic-prior residual campaign completed all six runs successfully; the scalar winner is the backward pointwise-control branch, but the campaign does not replace the current program winner.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14
Execution report	output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_execution_report.md
Leaderboard	output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_leaderboard.yaml
Best run pointer	output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_best_run.yaml
Manifest	output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_manifest.yaml
Planning report	doc/reports/campaign_plans/wave_3/2026-06-14-19-54-55_wave3_harmonic_prior_residual_campaign_plan_report.md
Technical package	doc/technical/2026-06/2026-06-14/2026-06-14-19-54-55_wave3_harmonic_prior_residual_campaign_package.md

Execution Summary

Field	Value
Campaign	wave3_harmonic_prior_residual_campaign_2026_06_14
Generated at	2026-06-15T15:30:20
Completed runs	6
Failed runs	0
Profiles	pointwise_control , smooth_l1_structured
Surfaces	global , Fw , Bw
Best scalar run	te_wave3_harmonic_prior_residual_pointwise_control_bw
Best scalar family	wave3_harmonic_prior_residual_pointwise_control_bw
Best scalar test MAE	0.003363
Best scalar test RMSE	0.003902

Field	Value
Program winner changed	no
Current program winner	te_periodic_gru_sequence_remote_Bw
TE Curve Verification Pipeline status	not run during normal closeout

Directional Branch Results

Surface	Candidate	Profile	Test MAE	Test RMSE	Val MAE
global	s11_global	s11	0.003403	0.003785	0.003633
Fw	pw_fw	pw	0.003382	0.003779	0.003315
Bw	pw_bw	pw	0.003363	0.003902	0.003634

Wave 5.1 Leaderboard

Candidate labels are aliases for the corresponding `te_wave3_harmonic_prior_residual_*` run names stored in the campaign leaderboard. All candidates have 7,283 trainable parameters.

Rank	Candidate	Surface	Test MAE	Test RMSE	Val MAE
1	pw_bw	Bw	0.003363	0.003902	0.003634
2	pw_fw	Fw	0.003382	0.003779	0.003315
3	s11_global	global	0.003403	0.003785	0.003633
4	s11_bw	Bw	0.003431	0.003953	0.003644
5	pw_global	global	0.003451	0.003851	0.003611
6	s11_fw	Fw	0.003527	0.003900	0.003310

Profile Comparison

Surface	Pointwise MAE	Smooth-L1 MAE	Better Profile	Relative Gap
global	0.003451	0.003403	s11	1.40%
Fw	0.003382	0.003527	pw	4.12%
Bw	0.003363	0.003431	pw	1.97%

Interpretation

Wave 5.1 validates the lightweight harmonic-prior residual implementation as a real trainable branch, with all six directional candidates completing cleanly and registering as implemented benchmarks. The best scalar result is backward only, while the best global result comes from the smooth_l1_structured profile.

The result is useful but not yet promotable. The best Wave 5.1 scalar test MAE of 0.003363 deg is weaker than the current program winner te_periodic_gru_sequence_remote_Bw at 0.002344 deg, and also weaker than the best recent Wave 4.3 mixture-density backward branch on scalar MAE. The main positive signal is architectural: the candidate has only 7,283 trainable parameters and keeps an explicit harmonic-prior residual structure.

The profile comparison does not support treating the robust SmoothL1 branch as a universal default. It helps on the global surface, but the plain pointwise-control profile wins on both direction-specific surfaces. The next decision should therefore be curve-first and direction-specific, not based on scalar training MAE alone.

Registry Effects

Registry Scope	Effect
Family registries	Six new wave3_harmonic_prior_residual_* family bests are registered.
Program registry	No program-best promotion; te_periodic_gru_sequence_remote_Bw remains current.
Active campaign state	Cleared by this closeout after report and PDF validation.
Master summary	Updated to show Wave 5.1 campaign closeout and separate TE Curve Verification Pipeline boundary.

TE Curve Verification Pipeline Boundary

Normal campaign closeout intentionally did not execute the heavy official TE Curve Verification Pipeline offline verification matrix. The next optional step is a separate TE Curve Verification Pipeline verification refresh for the six Wave 5.1 candidates, including collage and overlay reports, so the harmonic-prior residual branch can be compared on curve shape, offset, and direction-specific compensation behavior.

Closeout Decision

Accept the campaign as successfully closed at the training-campaign level. Do not promote a new program winner from scalar metrics. Plan official TE Curve Verification Pipeline verification as the next separate acceptance step before deciding whether Wave 5.1 should feed the later multi-head or Wave 5.2 integration roadmap.